Visual catalogue

of Kristals

Different knowledge structures deserve distinct shapes. Explore archetypes, situated examples, and one common visual grammar.

19 archetypes · 21 profiles · 10 visual patterns · 23 assemblies

The visual grammar

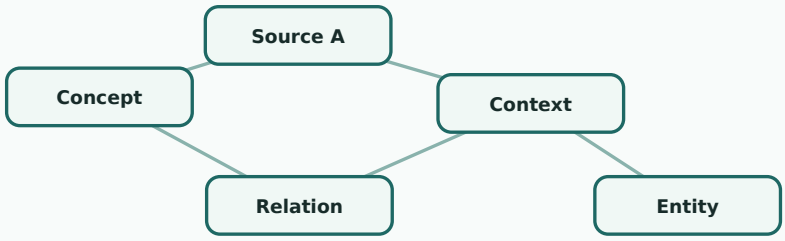
Linear	Ordered sequence of steps or transformations.
Branching sequence	Main process with decisions, exception branches and alternate outcomes.
Hub and satellites	A central case connects observations, proofs, state, and supporting documents.
Cluster / constellation	Interconnected clusters without one strict central hierarchy.
Tree / taxonomy	Branching classification from general groups to specific classes.
Timeline	Dated sequence of events, states, revisions or decisions.
Matrix / state table	Compare multiple items across multiple attributes or dimensions.
Casefile	A dossier connects records, notes, evaluations and next actions.
Multilayer	Several analytical layers or interpretations of a related subject.
Network / federation	Autonomous nodes linked by explicit references or coordination relationships.

Reading / integrity

The diagrams are fictional specimens, not evidence, Logical Contracts, or authorized decisions. Sources, access rights, and ownership boundaries belong to real Kristal instances.

ARK-01 Semantic Reference

Connected concepts, entities and sourced relationships. Reusable knowledge stores, dictionaries, educational and cultural references.



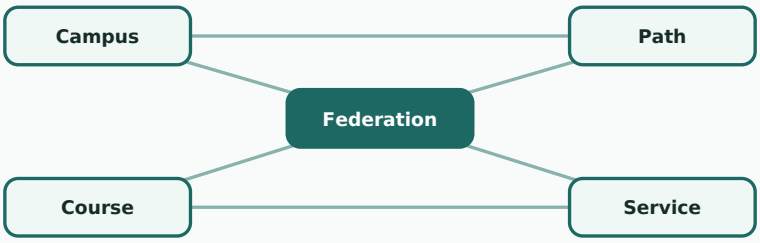
STRUCTURE Source-linked concepts, entities and typed relationships arranged in related thematic clusters.

USE CASES knowledge base · semantic dictionary · museum and education reference

Pattern: Semantic cluster

ARK-02 Federation / Topology

Hierarchy or network of separately identifiable knowledge states. Distributed campuses, organizational units and linked subsystems.



STRUCTURE Independent knowledge states linked by topology, scope, and separately owned parent-child references.

USE CASES multi-campus structure · federated sites · independent subsystems

Pattern: Federated network

ARK-03 Milestone and Proof Map

Sparse milestones, branches and pointers to proofs held elsewhere. Progress maps, mathematical roadmaps and cross-theory docking points.



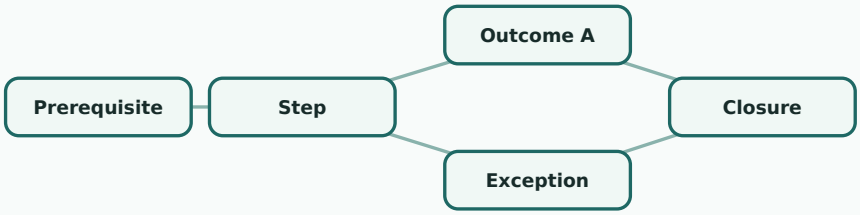
STRUCTURE Milestones and reached frontiers linked to proofs stored elsewhere; shortcuts do not replace proof.

USE CASES proof roadmap · research milestones · cross-theory references

Pattern: Sparse proof roadmap

ARK-04 Procedure / Protocol

Ordered and conditional steps, roles, prerequisites and exceptions. SOPs, institutional protocols and checklists.



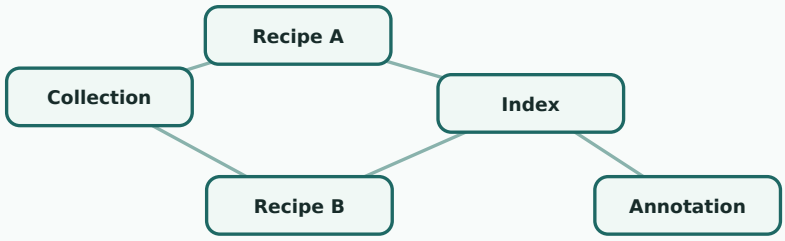
STRUCTURE Prerequisites lead to ordered actions, decisions, exception paths and authorized closure.

USE CASES standard operating procedure · field protocol · operations checklist

Pattern: Conditional protocol

ARK-05 Compilation / Collection

Curated, ordered and annotated independent works or knowledge items. Cookbooks, reference collections and anthologies.



STRUCTURE Independent entries collected, ordered and annotated without absorbing ownership of each entry.

USE CASES cookbook · anthology · curated reference

Pattern: Curated collection

ARK-06 Research Memory

Hypotheses, observations, trials and findings over time. Research notebooks, inquiry histories and lab studies.



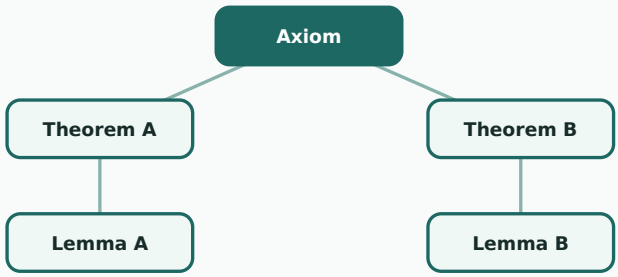
STRUCTURE Questions, hypotheses, trials and findings evolve over time with uncertainty preserved.

USE CASES research notebook · study record · experiment log

Pattern: Research history

ARK-07 Invariant / Formal Theory

Formal definitions, rules and logical dependencies. Theorem systems, invariant specifications and formal studies.



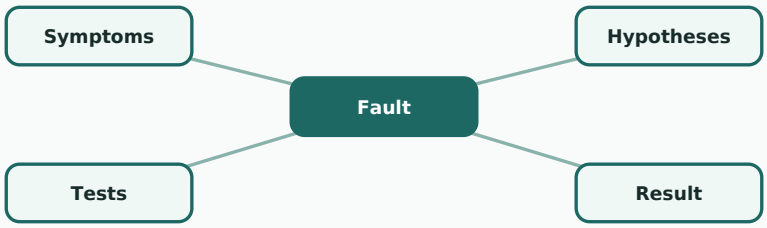
STRUCTURE Axioms and definitions support lemmas, theorems and explicit proof obligations.

USE CASES formal theory · invariant specification · proof dependency map

Pattern: Formal dependency tree

ARK-08 Causal Diagnosis

Symptoms, possible causes, test results and competing explanations. System troubleshooting, vehicle diagnosis and fault isolation.



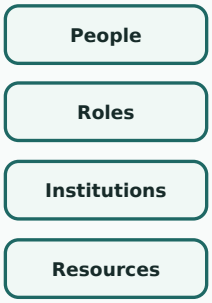
STRUCTURE A reported fault links observations, competing hypotheses, discriminating tests and results.

USE CASES system troubleshooting · vehicle diagnosis · root cause analysis

Pattern: Diagnostic hub

ARK-09 Multiplex / Organizational Model

Several relationship layers over shared entities. Organizational studies and social or institutional mappings.



STRUCTURE People, roles, institutions and resources are mapped in distinguishable relationship layers.

USE CASES organizational mapping · institutional studies · social relationships

Pattern: Organizational layers

ARK-10 Learning Path

Sequenced objectives, prerequisites and activities. Curricula and personalized learning journeys.



STRUCTURE Learning objectives, prerequisites, activities and assessments follow a navigable path.

USE CASES curriculum · guided training · learning plan

Pattern: Learning sequence

ARK-11 Event Ledger / Chronology

Dated observations, events and revisions. Incident logs, history, audit trails and case timelines.



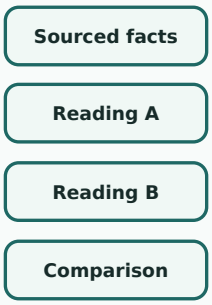
STRUCTURE Events, revisions, timestamps and provenance are recorded without assuming causality.

USE CASES incident timeline · audit trail · historical chronology

Pattern: Dated event sequence

ARK-12 Layered Interpretations

Separate derived readings of shared underlying material. Comparative analysis and alternative interpretations.



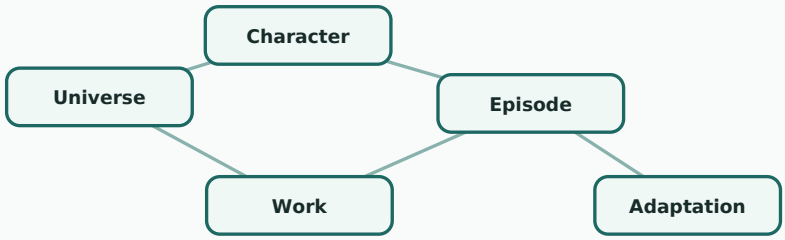
STRUCTURE Multiple readings and comparisons refer to the same source material without replacing it.

USE CASES comparative study · interpretation layers · analysis views

Pattern: Interpretation layers

ARK-13 Media and Creative Universes

Linked works, settings, characters and media manifestations. Worldbuilding, archives and media continuity.



STRUCTURE Works, characters, fictional events and media manifestations form a related universe.

USE CASES worldbuilding · media archive · creative continuity

Pattern: Creative universe cluster

ARK-14 Simulation / Transitions

State-transition definitions and hypothetical scenarios. Simulation, state machines and training situations.



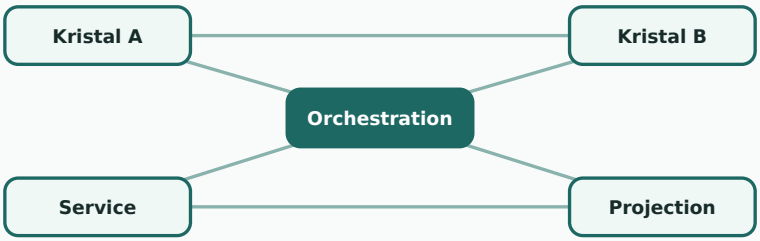
STRUCTURE Initial states progress through actions and transitions toward hypothetical outcomes.

USE CASES simulation · scenario analysis · state machine

Pattern: State transitions

ARK-15 Hybrid Composition

An assemblage of several kinds of independent knowledge structure. Cross-system models and multi-part applications.



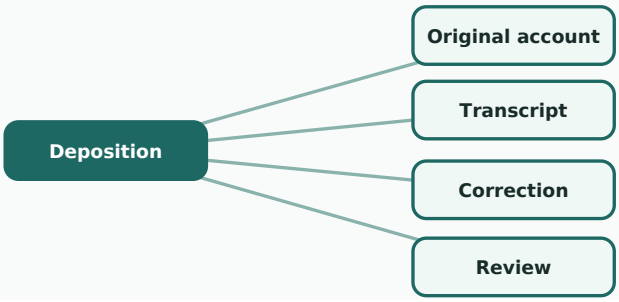
STRUCTURE Different knowledge-state forms interact through explicit references and interface boundaries.

USE CASES composite application · multi-system architecture · integrated knowledge

Pattern: Hybrid network

ARK-16 Testimony / Deposition

Attributed accounts with recordings, transcriptions and revisions. Witness records and declarative evidence dossiers.



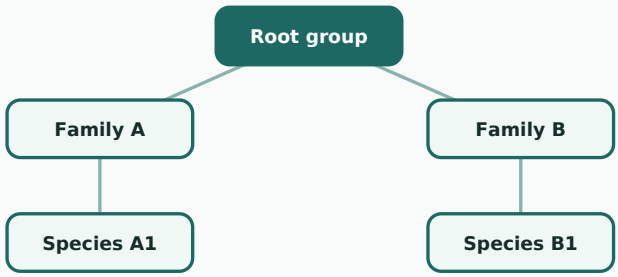
STRUCTURE Attributed accounts, recordings, transcriptions, corrections and independent evaluations remain distinct.

USE CASES witness statement · oral history · deposition record

Pattern: Testimony dossier

ARK-17 Taxonomy / Natural Classification

Classification tree with groups, subclasses and synonyms. Nature maps, biological categories and material typologies.



STRUCTURE Natural categories branch into families and species with criteria, synonyms and cited authorities.

USE CASES wildlife taxonomy · natural classification · specimen catalogue

Pattern: Classification tree

ARK-18 System-State Matrix

Observed condition of multiple systems across dimensions. Infrastructure status, asset state and configuration review.



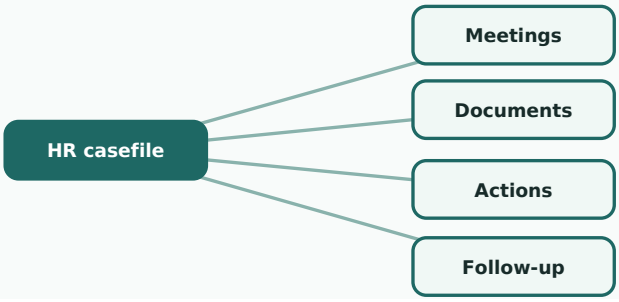
STRUCTURE Systems are measured on comparable dimensions at stated times, distinct from target configuration.

USE CASES infrastructure status · asset inventory · system monitoring

Pattern: System-state matrix

ARK-19 Casefile / Follow-Up

Central record of documents, notes, actions and follow-ups. Employee follow-up, case management and tracked interventions.



STRUCTURE A case centralizes meetings, documents, actions and assessments while keeping provenance and access controls.

USE CASES employee follow-up · HR casefile · case management

Pattern: Case tracking dossier

PRF-01 System Troubleshooting

Track symptoms, test competing hypotheses, and preserve observations while resolving technical incidents.



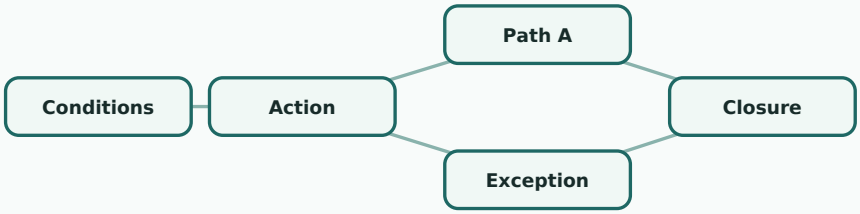
STRUCTURE An incident hub links reported symptoms, diagnostic tests, event logs, and observed system state.

USE CASES technical support · intermittent faults · failure analysis

Assembly: diagnostic-causal, registre-evenementiel, etat-systemes, composition-hybride

PRF-02 Procedural Environment

Document prerequisites, authorized roles, exception handling and conditional process steps.



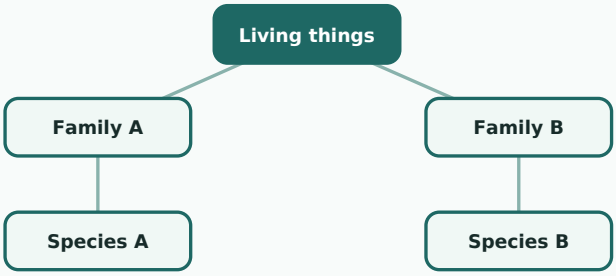
STRUCTURE A process branches at decision points; execution logs and simulations remain separate.

USE CASES medical protocols · police operations · industrial SOPs

Assembly: procedure-protocole, registre-evenementiel, simulation-transition

PRF-03 Nature Taxonomy

Classify living and non-living natural entities with explicit criteria and source references.



STRUCTURE A root category branches into groups, subgroups and specimens, with recorded synonyms.

USE CASES flora and fauna · minerals · natural objects

Assembly: taxonomie-classification, referentiel-semantique

PRF-04 System-State Tracking

Compare the observed condition of several systems across measurable dimensions.



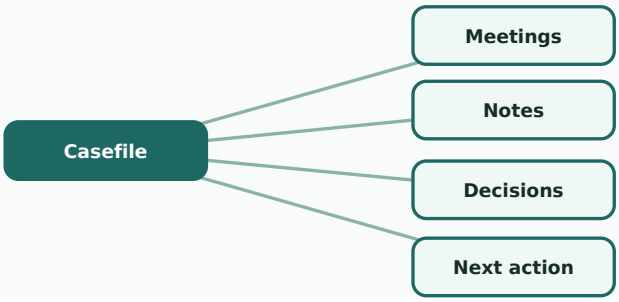
STRUCTURE Rows represent systems; columns represent observed state dimensions; the timeline is external.

USE CASES servers · personal devices · deployment states

Assembly: etat-systemes, registre-evenementiel

PRF-05 Employee Follow-Up

Structure a restricted employee casefile with meeting notes, actions and dated supporting records.



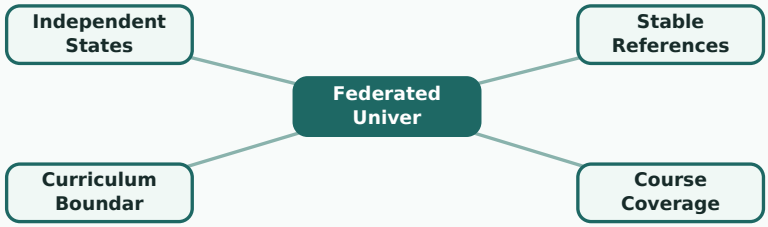
STRUCTURE A casefile links meetings, notes, decisions and separate testimony without collapsing their provenance.

USE CASES employee support · HR follow-up · case management

Assembly: dossier-suivi, registre-evenementiel, deposition-temoignage

PRF-06 Federated University / Campus

Federate separately owned institution, pathways and courses with stable links and explicit course boundaries.



STRUCTURE Institution references pathways; pathways reference independent course states.

USE CASES federated university / campus · education reference

Assembly: federation-topologie, parcours-pedagogique, referentiel-semantique

PRF-07 Moodle Learning Publication

Prepare educational content and mappings for external Moodle integration without automatic transfer or publication.



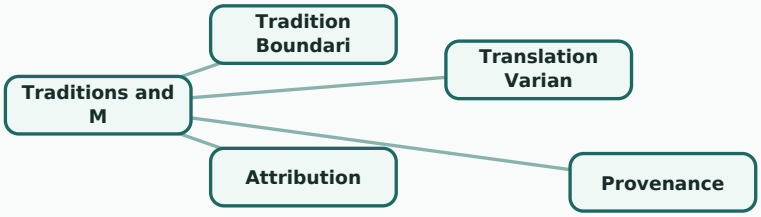
STRUCTURE Mapping of course identities, learning units and outcomes to LMS resources.

USE CASES moodle learning publication · education reference

Assembly: parcours-pedagogique, federation-topologie

PRF-08 Traditions and Mythology Collection

Collect and compare sourced traditions while preserving each community attribution and its own context.



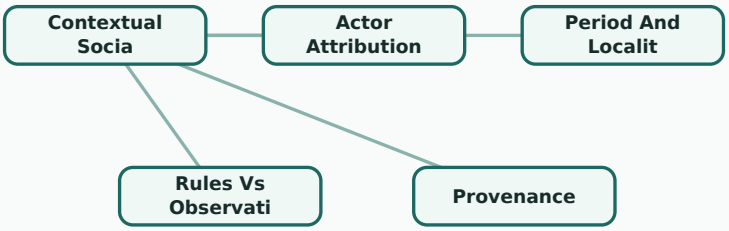
STRUCTURE Independent traditions are indexed and cross-referenced without collapsing distinct beliefs.

USE CASES traditions and mythology collection · culture reference

Assembly: compilation-collection, interpretations-couches, referentiel-semantique

PRF-09 Contextual Social Organizations

Map roles, responsibilities, resources and institutions across socially and historically situated contexts.



STRUCTURE Layers connect roles, institutions and resources over dated and sourced relationships.

USE CASES contextual social organizations · social-systems reference

Assembly: modele-multiplexe, registre-evenementiel, referentiel-semantique

PRF-10 Long-Proof Landmarks and Junctions

Navigate extensive mathematical proofs using endpoints, intermediate junctions, proof references and open frontiers.



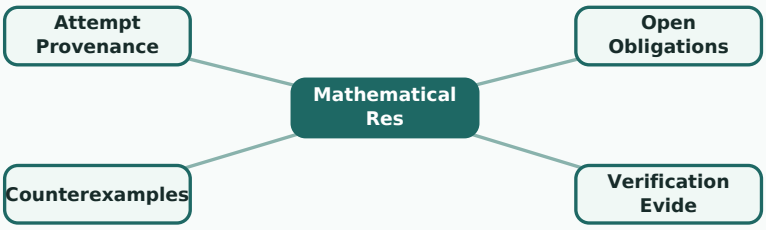
STRUCTURE A sparse route graph links proof landmarks and open areas, not formal proof derivations.

USE CASES long-proof landmarks and junctions · mathematics reference

Assembly: carte-jalons-preuves, invariant-theorie, memoire-recherche

PRF-11 Mathematical Research Frontier

Organize ongoing mathematical research, hypotheses, failed attempts and external verification routes.



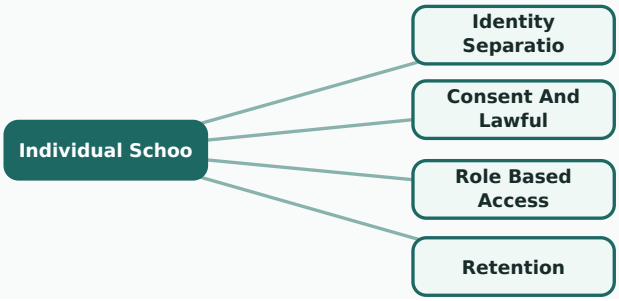
STRUCTURE Hypotheses and proof obligations branch into experiments and verification evidence.

USE CASES mathematical research frontier · mathematics reference

Assembly: memoire-recherche, carte-jalons-preuves, invariant-theorie

PRF-12 Individual School Follow-Up

Maintain an individually restricted educational record with changes, interventions and accountable decisions.



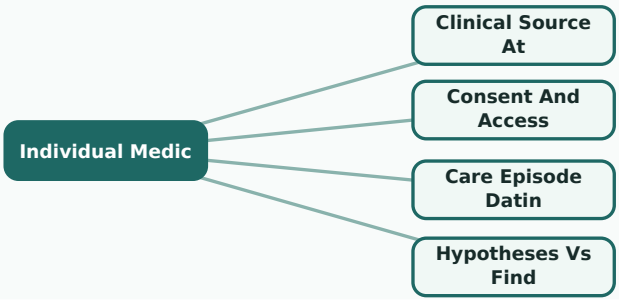
STRUCTURE Casefile links student history to assessments, supports, decisions and dates.

USE CASES individual school follow-up · education reference

Assembly: dossier-suivi, registre-evenementiel, parcours-pedagogique

PRF-13 Individual Medical Follow-Up

Organize longitudinal patient records and clinical events with strict source and access boundaries.



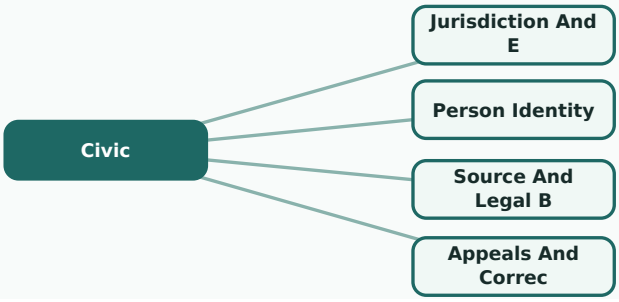
STRUCTURE Care episodes connect observations, decisions, documents and reviews without automated care delivery.

USE CASES individual medical follow-up · health reference

Assembly: dossier-suivi, registre-evenementiel, diagnostic-causal

PRF-14 Civic / Administrative Individual Casefile

Track administrative cases and documentary decisions over time under explicit jurisdiction and authority.



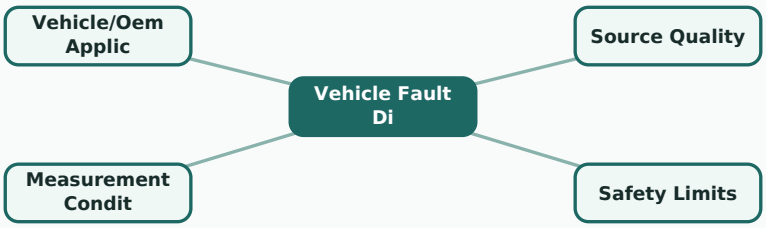
STRUCTURE A person-specific administrative case connects applications, notices, dates and decision records.

USE CASES civic / administrative individual casefile · civic reference

Assembly: dossier-suivi, registre-evenementiel, deposition-temoignage

PRF-15 Vehicle Fault Diagnosis

Diagnose vehicle failures using multiplex system topology, observations and testable hypotheses.



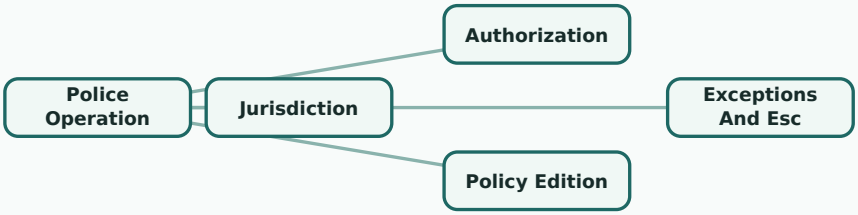
STRUCTURE Diagnostic hub ties symptoms and test points to causal candidate paths and observed states.

USE CASES vehicle fault diagnosis · automotive reference

Assembly: diagnostic-causal, modele-multiplexe, etat-systemes, registre-evenementiel

PRF-16 Police Operations Protocols

Organize versioned law-enforcement procedures, authorizations and exception pathways.



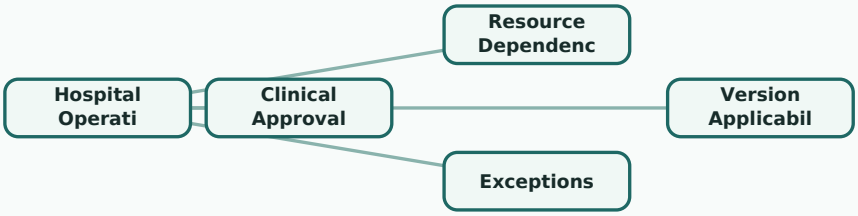
STRUCTURE A branching process identifies authorized roles, conditions and external event logging.

USE CASES police operations protocols · public-safety reference

Assembly: procedure-protocole, registre-evenementiel, simulation-transition

PRF-17 Hospital Operations Protocols

Structure institutional care workflows and operational procedures with exceptions, roles and audit.



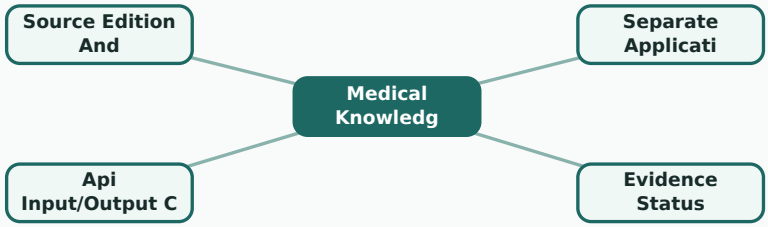
STRUCTURE Conditional workflow links roles, resources, escalation and external history.

USE CASES hospital operations protocols · health reference

Assembly: procedure-protocole, registre-evenementiel, etat-systemes

PRF-18 Medical Knowledge Orchestration

Design knowledge-state interfaces for independent clinical knowledge and diagnostic pathway software.



STRUCTURE Application orchestrates independent knowledge bases and adapters, with no inherited clinical authority.

USE CASES medical knowledge orchestration · health reference

Assembly: composition-hybride, referentiel-semantique, diagnostic-causal, procedure-protocole

PRF-19 School System and Rules Mapping

Map school sectors, cycles, transitions, regulations and temporally applicable conditions.



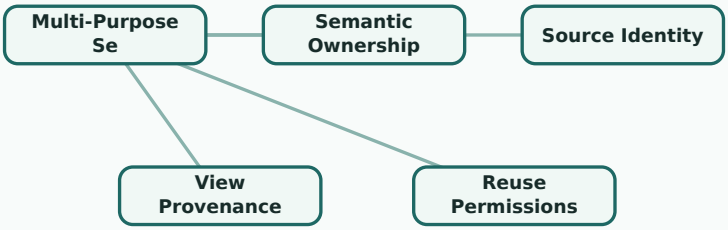
STRUCTURE Institutions, paths and rule layers connect through dated transitions and jurisdiction labels.

USE CASES school system and rules mapping · education reference

Assembly: modele-multiplexe, referentiel-semantique, parcours-pedagogique

PRF-20 Multi-Purpose Semantic Reference

Reuse a sourced semantic reference across teaching, museum, research and other readers.



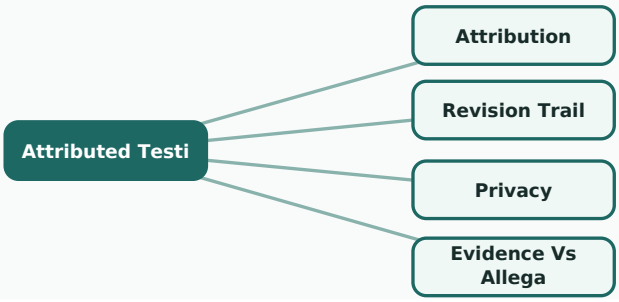
STRUCTURE One primary source reference feeds distinct projections without taking over each reader.

USE CASES multi-purpose semantic reference · cross-domain reference

Assembly: referentiel-semantique, parcours-pedagogique, interpretations-couches

PRF-21 Attributed Testimony and Inquiry

Document statements and accounts without treating their claims as established conclusions.



STRUCTURE Attributed statements attach to a restricted dossier and dated revisions.

USE CASES attributed testimony and inquiry · civic reference

Assembly: deposition-temoignage, dossier-suivi, registre-evenementiel